

EC3218 — Portfolio Analysis

Summer 2023 Examination · Paper 1 · Full Marking Scheme

Module	EC3218 — Portfolio Analysis
Internal Examiner	Dr. Fergal O'Connor
Institution	University College Cork
Session	Summer 2023, Paper 1
Duration	1.5 hours
Structure	Section A — Compulsory : Question A1 (50% of the paper). Section B — attempt either B1 or B2 (the other 50%). All questions carry equal weight within their section.
Materials	Non-programmable calculator permitted

Coverage note. A1 is compulsory, so this scheme covers it in full. Section B is answer **either** B1 **or** B2 — this scheme covers **both** in full so you can prepare either.

Own-figure rule. B1(a) is a multi-step numerical question — an answer that applies the correct method consistently from an earlier arithmetic slip still earns the marks available for the later steps.

Source basis. B2(c)'s discussion of gold's portfolio properties draws on Dr. O'Connor's own published research area (including He, O'Connor and Thijssen, 2018, on gold as a hedge/safe-haven asset, and Lucey and O'Connor, 2013, on bubbles in the gold price) — verified against the standard hedge/safe-haven/diversifier framework (Baur and Lucey, 2010; Baur and McDermott, 2010) since the specific gold material wasn't set out in the uploaded course pack.

Section A · Question 1 (Compulsory)

Market efficiency, financial bubbles, and a perfectly efficient world — 50% of the paper

A1(a)

[30 marks]

Explain what financial economists mean by "an efficient market". Use appropriate evidence in your answer.

A1(b)

[40 marks]

What is a Financial Bubble? Use an historical example in your answer.

A1(c)

[30 marks]

If all information were immediately and correctly incorporated into a stock's price as soon as the information was made public, what would be the only rational form of investment? What would be the role of financial analysts be in such a world?

FULL MODEL ANSWER

What financial economists mean by an efficient market: an efficient market is one in which asset prices **fully and rapidly reflect all available information** at any point in time — as Fama (1970) put it, prices "fully reflect" available information; Jensen's (1978) refinement adds an economic condition: prices reflect information up to the point where the marginal cost of acquiring and acting on further information would exceed the marginal benefit (profit) from doing so. The theory builds directly on the empirical observation that stock prices follow a **random walk** (Kendall, 1953) — price changes appear statistically independent from one day to the next, exactly what you'd expect if all predictable information were already priced in, leaving only genuinely unpredictable news to move prices further.

The three forms, and the evidence for each: market efficiency is not all-or-nothing — it comes in three nested forms, each making a stronger claim than the last, and the evidence for each differs in strength.

Form	Claim	Representative evidence
Weak	Prices reflect all past price/volume information	Generally well-supported, though seasonal anomalies (e.g. the January Effect, Rozeff and Kinney, 1976) provide some contrary evidence
Busse and Green (2002): stock prices adjusted to CNBC		

Semi-strong	Prices reflect all past AND public information	analyst mentions within seconds — broadly supportive, though anomalies like the P/E effect (Basu, 1977) and book-to-market effect (Fama and French, 1992) provide some contrary evidence
Strong	Prices reflect all information, public AND private	Evidence mostly rejects this form — Jaffe (1974) and others find corporate insiders and those following them earn genuine abnormal returns

The overall verdict: the weight of evidence suggests markets are usually weak-form efficient and mostly (if somewhat less consistently) semi-strong-form efficient, but are **not** strong-form efficient — a genuinely nuanced picture rather than a simple "markets are/aren't efficient" conclusion.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly defines an efficient market (Fama and/or Jensen's formulation) with the random walk foundation AND correctly distinguishes the three forms AND cites specific evidence for at least two forms AND gives the correct overall verdict (weak/semi-strong mostly supported, strong form mostly rejected).
15-22	15-22	Correct definition and forms given, with at least one specific piece of evidence cited.
8-14	8-14	Vague definition ("prices reflect information") without the three forms or specific evidence.
0-7	0-7	No credible definition given.

What a financial bubble is: a financial bubble occurs when an asset's price rises far beyond its intrinsic (fundamental) value — the value justified by the discounted stream of cash flows or utility the asset actually provides — and must, by its nature, eventually burst. As the module's own material puts it (Gurkaynak, 2008), a bubble exists when "investors are willing to pay more for the stock than they know is justified by the value of the discounted dividend stream."

Historical example — Tulip Mania (Holland, 1636-37): one of the earliest recorded economic bubbles. Tulip bulbs, newly fashionable in the Dutch Republic after their introduction from the Ottoman Empire in the 1550s, saw prices for rare varieties rise to extraordinary levels — at the peak, contracts for single bulbs reportedly traded for sums comparable to a skilled artisan's annual income or more. Prices collapsed abruptly in February 1637, when buyers at a routine bulb auction failed to appear, triggering a rapid, cascading collapse in prices across the market. Sir Isaac Newton's own much later lament after losing money in the 1720 South Sea Bubble — that he could predict the movement of celestial bodies but not the madness of markets — captures the same basic dynamic: even a sophisticated investor can be caught out once a bubble is underway.

EXAM TECHNIQUE

Other historical examples from the module's own material — the Mississippi Bubble (France, 1719-20), the South Sea Bubble (UK, 1720), 1920s US stocks up to the 1929 crash, and the dot-com bubble (1995-2001) — are equally creditable alternatives if a different example is preferred.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correct definition (price departing from, and eventually reverting to, fundamental value) AND a specific, correctly-described historical example with enough detail (dates, what happened, how it ended) to demonstrate genuine familiarity with the case, not just its name.
21-30	21-30	Correct definition with a historical example named but described only in general terms.
11-20	11-20	Vague definition without a historical example, or an example given with no definition.
0-10	0-10	No credible answer given.

The only rational form of investment in a perfectly, instantaneously efficient world: passive (index) investing. If every piece of information — past, public, and private — is already, instantly and correctly reflected in every asset's price the moment it becomes available, then no amount of analysis, research, or stock-picking skill can identify mispriced securities, since by construction there are none to find. In such a world, any attempt at active management — trying to select individual securities expected to outperform — cannot generate a genuine, sustainable risk-adjusted edge, and can only add cost (management fees, transaction costs, taxes from turnover) without any compensating benefit. The only rational response is to accept market prices as correct and hold a low-cost, fully diversified portfolio (in the CAPM sense, the tangency/market portfolio — see the EC3218 Summer 2025 scheme, Q. A2(c)) combined with the risk-free asset according to one's own risk tolerance, rather than paying for analysis that cannot, by the premise of the question, add any value.

The role of financial analysts in such a world — the Grossman-Stiglitz paradox: on the face of it, financial analysts would have **no useful role left to play** in security selection specifically, since their entire function — researching companies to identify mispriced securities — would be worthless where no mispricing can ever exist. But this immediately raises a deeper, well-known paradox (Grossman and Stiglitz, 1980): prices can only become *instantaneously and correctly* reflective of new information because analysts and other market participants spend real time and money *producing* that informational efficiency in the first place, by researching, trading on, and thereby impounding new information into prices. If analysts truly earned zero return for this costly effort — exactly what a perfectly efficient market implies — no one would have any incentive to do the analysis that makes markets efficient to begin with, which is **self-contradictory**: a market that is *too* efficient to reward information-gathering would stop being kept efficient at all. The practical resolution is that some degree of inefficiency must always persist, just enough to compensate analysts for their costly effort — meaning the premise of the question (perfect, instantaneous, costless efficiency) is itself an idealised limiting case rather than a literally achievable market state, and financial analysts' role in the real world is precisely to keep markets close to, without ever fully reaching, that limiting case.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies passive/index investing as the only rational strategy, with the reasoning (no mispricing to exploit, active management can only add cost) explained AND correctly identifies the Grossman-Stiglitz paradox — that perfect efficiency would remove the incentive to produce the very efficiency being assumed — as the resolution to the analyst's-

role question.

15-22	15-22	Correctly identifies passive investing as the only rational strategy with reasoning given, but the analyst's-role question is answered only with "they'd have no role" without the deeper paradox.
8-14	8-14	States passive investing is best without clearly explaining why, and no credible discussion of the analyst's role.
0-7	0-7	No credible answer given.

Question A1 Total: 100% (50% of the paper)

Section B · Question 1 (attempt EITHER B1 or B2)

APT arbitrage, portfolio performance measures, and the Fama-French model — 50% of the paper

B1(a)

[30 marks]

Arnor (A), Muppets Inc. (M) and Silmaril (S) are exposed to unexpected changes in the industrial production index (risk λ_1) and unexpected changes in interest rates (risk λ_2) as follows: $E(R_A) = \lambda_1 + 2\lambda_2$; $E(R_M) = -0.4\lambda_1 + \lambda_2$; $E(R_S) = 0.2\lambda_1 + \lambda_2$. Assume that over the next year both the market risk premium and the risk-free rate are 0, $\lambda_1 = 4\%$ and $\lambda_2 = 5\%$. Arnor and Muppets Inc. are both currently priced at 45p and Silmaril is 30p. What are the expected returns and expected prices of the stocks in one year's time, according to the APT? Your firm's research department confidently predicts that in one year's time the actual prices of Arnor, Muppets Inc. and Silmaril will be 50p, 46p and 35p. Arnor and Muppets Inc. thus appear to be currently slightly overvalued, and Silmaril is considerably undervalued. A new analyst suggests that you profit from this misvaluation by shorting one share each in Arnor and Muppets Inc. and going long three shares in Silmaril and close your positions in one year's time. What profit will you make, assuming the research department's predictions turn out to be correct?

B1(b)

[40 marks]

Discuss the Sharpe, Treynor, and Jensen measures of portfolio performance evaluation, and the situations in which each measure is the most appropriate measure.

B1(c)

[30 marks]

Summarise how the Fama and French 3-factor model works, using an equation. Do you think that the three factors included make up a logical explanation of

how share returns may be explained?

FULL MODEL ANSWER

Step 1 — expected returns under the APT: with $\lambda_1 = 4\%$ and $\lambda_2 = 5\%$, and the market risk premium and risk-free rate both zero (so these two factor risk premiums are the entire story), substitute directly into each stock's factor equation.

Stock	Formula	Expected return
Arnor	$\lambda_1 + 2\lambda_2 = 4\% + 2(5\%)$	14.0%
Muppets Inc.	$-0.4\lambda_1 + \lambda_2 = -0.4(4\%) + 5\%$	3.4%
Silmaril	$0.2\lambda_1 + \lambda_2 = 0.2(4\%) + 5\%$	5.8%

Step 2 — expected prices in one year (APT-implied, fair value): apply each expected return to the current price.

Stock	Calculation	Expected (APT-fair) price in 1 year
Arnor	$45p \times 1.140$	51.30p
Muppets Inc.	$45p \times 1.034$	46.53p
Silmaril	$30p \times 1.058$	31.74p

Step 3 — why the research department's predictions imply mispricing: the research department's *actual predicted* prices (50p, 46p, 35p) differ from the APT-fair prices above. Arnor's predicted 50p is *below* its APT-fair 51.30p, and Muppets' predicted 46p is *below* its APT-fair 46.53p — both stocks are predicted to earn *less* than the return the APT says they should earn for their risk, meaning their current 45p prices are **too high** (overvalued) relative to fair value. Silmaril's predicted 35p is *above* its APT-fair 31.74p — it's predicted to earn *more* than its required return, meaning its current 30p price is **too low** (undervalued) — exactly matching the question's own framing.

Step 4 — the arbitrage: short 1 Arnor, short 1 Muppets, long 3 Silmaril: since Arnor and Muppets are overvalued, the profitable position is to **sell them short** (borrow and sell now, buy back later — profiting if the price falls or underperforms); since Silmaril is undervalued, the profitable position is to **go long** (buy now, hoping to sell at a relatively higher price later).

Cash flow today	Calculation	Amount
Short 1 Arnor (receive proceeds)	+45p	+45.00p
Short 1 Muppets (receive proceeds)	+45p	+45.00p
Buy 3 Silmaril (pay cost)	$-3 \times 30p$	-90.00p
Net cash flow TODAY	45 + 45 - 90	0.00p

A genuinely zero-cost position: notice the position requires **no net investment at all today** — the short-sale proceeds from Arnor and Muppets exactly fund the purchase of the three Silmaril shares. This is the hallmark of a true arbitrage: any profit earned is pure, riskless profit relative to the predicted prices, not compensation for capital tied up.

Cash flow in 1 year (at the research department's predicted prices)	Calculation	Amount
Buy back 1 Arnor to close the short	−50p	−50.00p
Buy back 1 Muppets to close the short	−46p	−46.00p
Sell 3 Silmaril	+3 × 35p	+105.00p
Net cash flow IN 1 YEAR	−50 − 46 + 105	+9.00p

Result: the strategy generates a **profit of 9p** per unit of the position (1 Arnor short + 1 Muppets short + 3 Silmaril long), realised entirely in one year's time, on an initial net investment of exactly zero — assuming, as stated, that the research department's predicted prices turn out to be correct.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	All three expected returns correctly calculated (14.0%, 3.4%, 5.8%) AND all three APT-fair prices correctly calculated (51.30p, 46.53p, 31.74p) AND the zero-cost property of the position correctly identified AND the final profit of 9p correctly calculated.
15-22	15-22	Expected returns and APT-fair prices correctly calculated, but an error in setting up or calculating the final arbitrage cash flows.
8-14	8-14	Expected returns correctly calculated, but the APT-fair prices or the arbitrage cash flow structure are incorrect.
0-7	0-7	No coherent APT calculation attempted.

Sharpe Ratio: $(\bar{r}_p - \bar{r}_f)/\sigma_p$ — average portfolio excess return divided by the standard deviation of returns. It measures return per unit of **total** risk, and is the slope of the portfolio's own Capital Allocation Line. **Most appropriate when the portfolio represents an investor's entire risky holding**, since in that case total risk (not just the systematic portion) is what actually matters to the investor's overall wellbeing.

Treynor Measure: $(\bar{r}_p - \bar{r}_f)/\beta_p$ — average portfolio excess return divided by portfolio beta. It measures excess return per unit of **systematic** risk only, on the reasoning that idiosyncratic risk should already be diversified away in a well-constructed portfolio. **Most appropriate when the portfolio is one of several sub-portfolios** combined into a larger, well-diversified fund, since in that context only the portfolio's systematic risk contributes to the overall fund's risk.

Jensen's Alpha: unlike the two ratio measures above, alpha is not a ratio — it is the portfolio's actual average return minus its CAPM-expected return: $\alpha_p = \bar{r}_p - [\bar{r}_f + \beta_p(\bar{r}_m - \bar{r}_f)]$. A positive alpha indicates the portfolio outperformed what its systematic risk alone would predict. **Most appropriate for the same "one of many sub-portfolios" context as the Treynor measure** — but, unlike Treynor, alpha is expressed in absolute return terms (e.g. "2% per year") rather than as a ratio, which makes it easier to interpret directly as a euro/percentage amount of value added, and easier to aggregate meaningfully across multiple sub-portfolios of different sizes and risk levels within a larger fund.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	All three measures correctly defined with correct formulas AND each correctly distinguished by which type of risk it measures return against (total vs. systematic) AND the appropriate use-case for each is correctly explained (entire holding → Sharpe; one of several sub-portfolios → Treynor/Jensen).
21-30	21-30	All three measures correctly defined, but the appropriate use-case distinctions are missing or only partially given.
11-20	11-20	Only two of the three measures correctly defined.
0-10	0-10	Fewer than two measures correctly defined.

How the model works: the Fama-French 3-factor model explains a portfolio's returns using three factors rather than CAPM's single beta:

Equation: $R_p - R_f = a + b_1(R_m - R_f) + b_2 \text{SMB} + b_3 \text{HML} + e$

- **(R_m – R_f):** the market factor, CAPM's own market risk premium.
- **SMB (Small Minus Big):** the excess return of small-cap stocks over large-cap stocks.
- **HML (High Minus Low):** the excess return of high book-to-market ("value") stocks over low book-to-market ("growth") stocks.

Opinion: do the three factors make up a logical explanation?

The case that they do: each factor has a plausible economic story behind it, not just an empirically-fitted coefficient. The market factor captures broad, undiversifiable economic risk, exactly as CAPM predicts. Small firms plausibly carry **genuinely higher risk** — thinner analyst coverage, less access to capital markets, higher failure rates, more exposure to financial distress in downturns — which is a coherent risk-based story for why SMB should command a positive premium. High book-to-market (value) firms are often value firms precisely *because* the market has marked them down due to distress or poor prospects, so a value premium is similarly consistent with compensation for distress risk.

The case that they don't (or not fully): the module's own material notes real criticisms: the model still lacks a fully worked-out **theoretical** foundation (unlike CAPM, which is derived from an equilibrium model, SMB and HML were originally identified *empirically*, then a risk story was fitted around them afterward) — and Lakonishok, Shleifer and Vishny (1994) directly challenge the risk-based story, finding value stocks did *not* underperform in "bad" economic states the way a genuine risk premium should predict, which is more consistent with behavioural mispricing (see the EC3218 Summer 2025 scheme, Q. A1(c)) than with SMB/HML representing real, priced risk. The model has also been found to work considerably less well outside the US — Gregory, Tharyan and Christidis (2013) find the factors are "not consistently and reliably priced" in the UK context specifically.

A balanced personal view: the most defensible position is that the three factors are a **reasonable, if imperfect and empirically-motivated rather than purely theory-derived**, explanation — genuinely useful for performance attribution and better than CAPM's single factor at explaining the cross-section of returns, but the underlying debate over whether SMB and HML represent real priced risk or exploitable behavioural mispricing (Fama-French's own view vs. LSV's) remains genuinely unresolved rather than settled in the model's favour.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly states the full equation with all three factors defined AND gives a genuine, reasoned opinion engaging both the case for (plausible risk stories for each factor) and the case against (empirical rather than theoretical origin, LSV's evidence against a pure risk explanation, weaker performance outside the US).
15-22	15-22	Correct equation given with a one-sided opinion (only the case for, or only the case against).
8-14	8-14	Equation stated with factors named, but no genuine opinion/reasoning given.
0-7	0-7	No credible equation or discussion given.

Question B1 Total: 100% (50% of the paper)

Full Marking Scheme · Study-Uni

Section B · B1

EC3218 · Portfolio Analysis

Summer 2023 Examination · Paper 1

Section B · Question 2 (attempt EITHER B1 or B2)

The efficient frontier, active vs. passive investing, and gold in a portfolio — 50% of the paper

B2(a)

[30 marks]

Explain with diagram(s) and in your own words the "Efficient Frontier".

B2(b)

[40 marks]

In your opinion should investors hold a passive or an active portfolio?

B2(c)

[30 marks]

When adding gold to an asset allocation program that currently includes only stocks, bonds and cash, which of the properties of gold returns affect portfolio risk? Explain your answer.

FULL MODEL ANSWER

In your own words: the **Minimum Variance Frontier** plots, for every achievable level of expected portfolio return, the smallest possible risk (standard deviation) that combination of risky assets can achieve. The **efficient frontier** is the **upper half** of this frontier — the part running from the single lowest-risk point (the **minimum variance portfolio**) upward —

since for every point on the *lower* half, there's a portfolio directly above it on the upper half offering the exact same risk but a higher expected return, making the lower half strictly worse for any rational, return-seeking investor.

Minimum Variance Frontier, Efficient Frontier, and the CAL

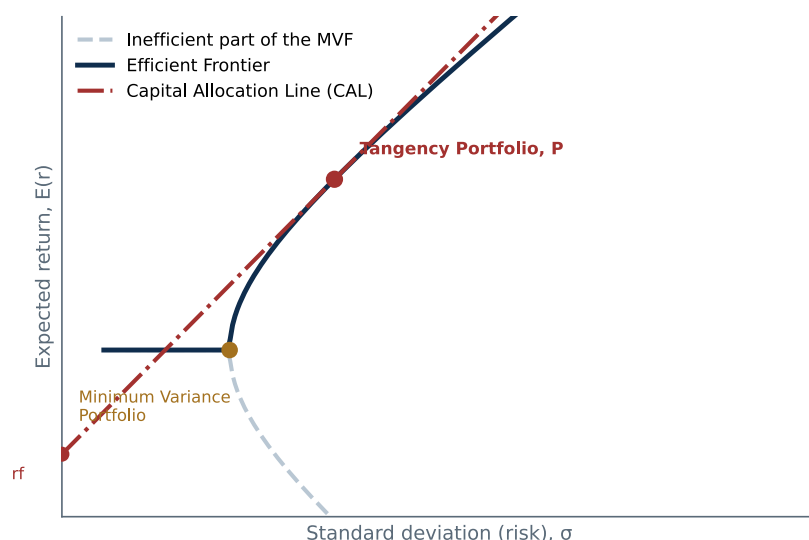


Fig. 1 — The Minimum Variance Frontier (both the efficient upper arc and the inefficient lower arc), the minimum variance portfolio, and the Capital Allocation Line tangent to the frontier at the tangency portfolio, P .

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correct definition of the MVF and the efficient frontier as its upper arc AND the diagram correctly shows both, with the minimum variance portfolio marked AND the dominance reasoning for why the lower arc is inefficient is explained in the student's own words.
15-22	15-22	Correct diagram given, but the dominance reasoning is only asserted, not explained.
8-14	8-14	Diagram attempted with some correct elements, but the efficient frontier/MVF distinction is unclear.
0-7	0-7	No credible diagram or explanation given.

The case for passive investing: the weight of empirical evidence on market efficiency (see Q. A1(a)) suggests markets are at least reasonably efficient at the weak and semi-strong levels, meaning consistently identifying mispriced securities through analysis is genuinely difficult. Layered on top of this, active management carries real, certain costs — higher fees, more frequent trading (transaction costs, tax drag from turnover) — that a passive, low-cost index fund avoids entirely. Even where genuine manager skill exists, research on performance persistence finds that much of what looks like skill is actually explained by factor exposure (e.g. momentum, size, value tilts) or differences in expense ratios rather than true, repeatable security-selection ability (Carhart, 1997), and that apparent "winning" managers are hard to identify reliably in advance.

The case for active investing: markets are not perfectly efficient — genuine anomalies exist (Q. A1(a)), some managers do show evidence of skill in specific, narrower dimensions (e.g. Busse's 1999 finding of successful volatility timing among some mutual fund managers), and active management retains value in contexts a pure index fund can't address — for example, tailoring a portfolio to an individual's specific tax situation, liquidity needs, ESG preferences, or risk constraints that a broad index doesn't accommodate.

A reasoned opinion: the most defensible position, on the weight of the evidence, is that a **core passive allocation, with active management reserved for a smaller satellite portion** (if used at all) is the more rational default for most investors — the evidence that active management reliably beats a low-cost index fund net of fees is weak and inconsistent, while the cost savings and simplicity of a passive approach are certain. This isn't an argument that active management is never worthwhile — specific, narrow, well-evidenced skills (like the volatility timing Busse documents) can add genuine value — but the burden of proof should sit with the active strategy to demonstrate it clears its own costs persistently, not with the passive default to justify itself.

EXAM TECHNIQUE

This is a genuinely open, opinion-based question — a well-reasoned case for a more active approach can score equally well, provided it directly engages with the efficiency evidence and the cost/persistence problem, rather than ignoring them.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Presents a genuine, reasoned opinion AND engages both the case for passive (efficiency evidence, costs, persistence problems) and the case for active (specific anomalies, narrow evidence of skill, individual tailoring needs) AND the opinion is clearly justified by weighing the two rather than asserted.
21-30	21-30	A clear opinion given engaging one side of the debate in depth, with the other side only briefly acknowledged.
11-20	11-20	An opinion stated without substantive engagement with the underlying evidence on either side.
0-10	0-10	No credible opinion or reasoning given.

EXAM TECHNIQUE

This question wasn't set out in the uploaded course material; the discussion below follows the standard hedge/safe-haven/diversifier framework (Baur and Lucey, 2010; Baur and McDermott, 2010), including He, O'Connor and Thijssen's (2018) own published findings on gold's properties as a hedge and safe-haven asset for equities.

The relevant properties, and the standard definitions: what matters for portfolio risk isn't just an asset's own volatility, but its **correlation** with the assets already in the portfolio — a highly volatile asset can still reduce overall portfolio risk if its returns move independently of, or opposite to, the rest of the portfolio.

- **Hedge property:** defined (Baur and Lucey, 2010) as being uncorrelated or negatively correlated with another asset *on average*, across all market conditions. Gold has been found to act as a hedge against stocks on average in several major markets (the US, UK), meaning adding gold to a stock/bond/cash portfolio can reduce overall portfolio variance even in ordinary market conditions, simply because gold's returns aren't driven by the same underlying economic factors as equities.
- **Safe-haven property:** a distinct, stronger property — being uncorrelated or negatively correlated with another asset *specifically during periods of market stress or crashes* (Baur and Lucey, 2010). This is the property most directly relevant to reducing tail risk: a diversifier that only helps in calm markets is far less valuable than one that helps precisely when the rest of the portfolio is falling sharply. Evidence (including He, O'Connor and Thijssen, 2018) supports gold acting as a safe haven during extreme stock market downturns in several major markets, though Baur and Lucey (2010) find this safe-haven property tends to be

short-lived rather than permanent.

- **Gold's own volatility:** gold is not a low-volatility asset in absolute terms — its price can move substantially on its own. This matters because the *portfolio* risk-reduction benefit of gold's low/negative correlation has to be weighed against the risk gold itself contributes as a standalone position; a genuinely effective diversifier needs both low correlation with existing holdings *and* a sensible position size, since even a zero-correlation asset can raise portfolio variance if allocated too large a share, given its own non-trivial volatility.
- **No income yield:** unlike stocks (dividends), bonds (coupons), or cash (interest), gold generates no cash flow of its own — its return comes entirely from price appreciation. This is a genuine opportunity cost of holding gold rather than a source of portfolio *risk* as such, but it does mean gold's diversification benefit (lower correlation, safe-haven behaviour in crises) has to be weighed against forgone income the other three asset classes would have provided.

Putting it together: adding gold to a stocks/bonds/cash portfolio primarily affects portfolio risk through its **low-to-negative correlation with equities (and, to varying degrees, bonds)**, both on average (the hedge property) and especially during market crises (the safe-haven property) — this is what can genuinely lower overall portfolio variance, particularly tail risk during equity market crashes. The extent of that benefit, however, is bounded by gold's own non-trivial volatility as a standalone asset and by the fact that its safe-haven property, while real, has been found empirically to fade relatively quickly after a crisis event rather than persisting indefinitely.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies low/negative correlation with stocks as the key risk-reducing property AND correctly distinguishes the hedge property (on average) from the safe-haven property (specifically during crises) AND correctly notes gold's own volatility and lack of yield as countervailing considerations.
15-22	15-22	Correctly identifies low/negative correlation as the key property, but the hedge/safe-haven distinction or the countervailing considerations are missing.
8-14	8-14	Vague reference to gold being "a good diversifier" without explaining the correlation mechanism.
0-7	0-7	No credible answer given.

Question B2 Total: 100% (50% of the paper)